

AI Career Compass : A Data-Driven Intelligent System For Personalized Career Guidance

Aditya Raj Pathak¹, Avinash Singh¹, Aditya Parmar

¹School of Computer Science and Engineering, Galgotias University
Greater Noida, Uttar Pradesh, India — 201310

ABSTRACT

Keywords: *Career Recommendation, Machine Learning, Natural Language Processing, TF-IDF, Logistic Regression, FastAPI, NIRF Rankings*

Career decision-making is one of the most consequential challenges faced by undergraduate students in India, where the mismatch between graduate skill sets and industry requirements continues to widen. Traditional career counselling remains inaccessible to the majority of students due to resource constraints, geographic limitations, and the inability of manual advisory systems to scale effectively. This paper presents an AI-Based Career Guidance System that leverages supervised machine learning and natural language processing to deliver personalised, data-driven career recommendations. The system is trained on a consolidated dataset of 741 labelled records derived from two publicly available sources: an AI-based career recommendation dataset and a career recommender survey dataset, each preprocessed and merged through a rigorous data engineering pipeline. Three machine learning classifiers — Random Forest, Logistic Regression, and Linear Support Vector Classification — were trained on Term Frequency-Inverse Document Frequency (TF-IDF) vectorised representations of students' self-reported skills and interests. Experimental evaluation demonstrates that Logistic Regression achieves the highest classification performance, attaining a Top-1 accuracy of 45.3%, a Top-3 accuracy of 65.1%, and a Top-5 accuracy of 77.9% across 22 consolidated career categories. These results are contextualized against a random-chance baseline of 4.5%, confirming a tenfold improvement. The system further integrates National Institutional Ranking Framework (NIRF) 2024 data for personalised college recommendations and a curated fresher job listings dataset for employment matching, filtered by the student's academic scores, budget, preferred city, and study mode. The back-end REST API is deployed via FastAPI and Uvicorn, providing sub-50 millisecond response times. The proposed system demonstrates the viability of lightweight machine learning architectures for educational recommendation in resource-constrained environments and establishes a reproducible baseline for future deep learning extensions.

1. INTRODUCTION

1.1 Background

Career planning is a multifaceted process that demands an accurate understanding of one's own competencies, the evolving demands of the labour market, and the landscape of available educational pathways. In India, where over 7.5 million students graduate from undergraduate programmes annually, the gap between graduate qualifications and industry expectations has been identified as a significant contributor to structural unemployment. The National Employability Report (Aspiring Minds, 2019) estimated that only 4.77% of engineering graduates in India possessed the requisite skills for software-related roles, while broader skill mismatches persisted across disciplines including commerce, science, and the humanities.

The emergence of Artificial Intelligence and Machine Learning in the domain of educational technology has created new possibilities for intelligent, scalable, and personalised guidance systems. Unlike rule-based expert systems that rely on predefined heuristics, machine learning models can generalise from historical data to capture nuanced relationships between individual skill profiles and successful career outcomes. Techniques such as Natural Language Processing (NLP), specifically vector space models built on TF-IDF weighting, enable the transformation of free-form textual skill descriptions into quantitative representations amenable to statistical classification.

This research situates itself at the intersection of educational data mining, NLP-based text classification, and career information systems. The proposed system ingests a student's self-described skills and interests, processes them through a trained classification pipeline, and returns ranked career recommendations accompanied by matched college institutions sourced from the NIRF 2024 rankings and matched fresher employment opportunities from a curated job listings corpus. The entire inference stack is exposed through a RESTful API that enables integration with any front-end interface.

1.2 Literature Review

Early computational approaches to career counselling employed rule-based expert systems that codified the decision logic of human career counsellors. Holland's RIASEC (Realistic, Investigative, Artistic, Social, Enterprising, Conventional) typology formed the theoretical basis for many such systems, and its influence persists in contemporary psychometric assessment tools. However, these systems lacked the adaptability to respond to the rapid evolution of new job roles driven by the digital economy.

Subsequent work explored collaborative filtering techniques analogous to those used in recommender systems for e-commerce. Reschly and Christenson (2006) demonstrated that peer-performance data could serve as a useful proxy for career aptitude, while more recent work by Gupta et al. (2020) applied matrix factorisation methods to construct student-career affinity matrices from historical placement records. A significant limitation of collaborative filtering in this domain is the cold-start problem: new students have no interaction history, making similarity-based recommendations impossible.

The application of supervised text classification to career prediction gained traction with the availability of large-scale survey datasets. Kumar et al. (2021) employed a Naive Bayes classifier on a multi-label career prediction task, achieving 52% accuracy on a binary formulation of the problem. Jain and Sharma (2022) extended this work using Support Vector Machines with TF-IDF features across 15 career categories, reporting a macro-averaged F1 score of 0.48. Deep learning approaches, including LSTM-based sequence classifiers and BERT fine-tuned on career-description corpora, have demonstrated superior performance (Nguyen et al., 2023), though their computational requirements remain a barrier to deployment in low-resource settings.

Integrated career guidance platforms that combine prediction with institutional matching remain underexplored in the literature. Roy and Banerjee (2022) proposed a hybrid system that paired skill-based prediction with a hand-curated list of institutions, but relied on static mappings that did not account for budget,

location, or programme-mode preferences. The present work addresses this gap by incorporating dynamic, multi-attribute filtering against government-standardised NIRF data.

1.3 Research Gap

A review of existing literature reveals three principal gaps that the present research seeks to address. First, most career prediction systems operate as isolated classifiers without providing actionable downstream guidance such as institution matching or employment opportunity discovery, limiting their practical utility for end users. Second, the majority of published systems do not account for multi-attribute student constraints — specifically academic performance thresholds, geographical preferences, tuition budget, and study modality — when generating institutional recommendations. Third, prior work on lightweight, deployable career guidance systems lacks a transparent evaluation framework that benchmarks model performance against a probabilistic baseline appropriate to the number of career categories, making it difficult to contextualise reported accuracy figures.

Additionally, no published system has employed a fully open, reproducible pipeline — from raw CSV ingestion through preprocessing, model training, and REST API deployment — accompanied by per-class classification diagnostics. The present work contributes precisely such a pipeline, including a detailed performance analysis stratified by career category that highlights classes where data sparsity limits predictive reliability.

1.4 Objective

The primary objective of this research is to design, implement, and evaluate a complete, end-to-end AI-Based Career Guidance System that accepts a student's natural-language skill and interest profile and returns personalised career, institution, and employment recommendations. The specific sub-objectives are:

- To construct a high-quality training corpus by merging and preprocessing two publicly available career datasets.
- To evaluate three supervised machine learning classifiers — Random Forest, Logistic Regression, and Linear SVC — under identical experimental conditions and select the best-performing model.
- To integrate NIRF 2024 college data and a fresher job listings dataset into the recommendation pipeline, enabling multi-attribute filtering.
- To deploy the system as a production-grade REST API with sub-50 millisecond response times, suitable for integration with any front-end client.

1.5 Scope and Constraints

The system is scoped to career prediction within the Indian higher-education and employment context. All institutional data is sourced from the NIRF 2024 rankings, restricting recommendations to NIRF-listed institutions. The career taxonomy is fixed at 22 consolidated categories derived from the training data; careers outside this taxonomy cannot be predicted. The classifier is trained on English-language free-text inputs; students who describe their skills in regional languages or code-switched text may experience degraded performance. Academic-score-based eligibility tagging employs heuristic thresholds based on NIRF score bands and is not intended as a formal admissions prediction tool. The system does not incorporate real-time labour market data; job listings are sourced from a static dataset and may not reflect current market conditions.

2. MATERIALS AND METHODS

2.1 Materials Used

Four primary datasets were employed in the construction of the system. Table 1 summarises their characteristics prior to preprocessing.

Table 1: Summary of Raw Datasets

Dataset Name	Rows	Columns	Description
AIbased_Career_Recommendation_System.csv	200	8	Synthetic career recommendation records with Skills, Interests, and Recommended_Career labels
career_recommender.csv	1,195	12	Survey responses from real working professionals recording UG course, skills, interests, CGPA, and job title
NIRF_Ranking_2024.csv	470	13	National Institutional Ranking Framework 2024 data including institution name, city, state, field, and composite score
fresher_jobs.csv	845	3	Fresher-level job listings with Job Title, Location, and Salary fields sourced from Indian job portals

The training corpus was constructed by combining the first two datasets after preprocessing. The NIRF and job datasets served exclusively as lookup tables for downstream recommendation filtering, contributing no labelled samples to the classification training process.

2.2 Methodology / Procedure

The research followed a structured pipeline comprising three phases: data preprocessing, model training and selection, and REST API deployment. Each phase is described below.

Phase 1 — Data Preprocessing. Raw datasets were loaded using the pandas library. From the AI career dataset, only the Skills, Interests, and Recommended_Career columns were retained, and rows with missing values in these fields were dropped. Text was normalised to lowercase and stripped of leading and trailing whitespace. From the career recommender survey, question-style column headers were renamed, and rows corresponding to students, unemployed respondents, and inapplicable entries were excluded, retaining only records from working professionals. Inconsistent skill separators (semicolons) were standardised to commas. The two cleaned datasets were concatenated, deduplicated on the joint key (Skills, Interests, Recommended_Career), and filtered to remove records with implausibly short skill or interest strings (three characters or fewer).

Phase 2 — Career Label Consolidation. The raw datasets contained 381 unique career label strings owing to free-text entry by survey respondents. These were manually consolidated into 22 semantically coherent career categories (e.g., merging 'Software Engineer', 'Backend Developer', 'Java Developer', and 'Programmer' into 'Software Developer'). Career classes with fewer than eight training examples were excluded to prevent the classifier from learning unreliable patterns from sparse data. This yielded a final training corpus of 344 records spanning 22 categories.

Phase 3 — NIRF Data Preprocessing. The NIRF dataset was cleaned by removing a spurious index column, renaming cryptic column identifiers (e.g., TLR, RPC, GO) to descriptive equivalents, and mapping institution-specific field names (Engineering, Management, Medical, etc.) to the broader career domain taxonomy used by the classifier. Institutions were sorted by ascending NIRF rank to ensure top-ranked results appear first in recommendations.

Phase 4 — Job Dataset Preprocessing. The fresher jobs dataset presented significant challenges in the Salary field, which contained values in multiple heterogeneous formats including monthly ranges, annual ranges, single-sided bounds, and 'Not specified' entries. A custom parse_salary() function was implemented using

Python regular expressions to extract all numerical tokens from each salary string, average range values, and convert monthly figures to annual equivalents by multiplication by twelve. City names were extracted from full address strings by splitting on commas and retaining the first token.

2.3 Tools and Techniques

The system was implemented entirely in Python 3.11. The following libraries constituted the core toolchain:

- pandas (v2.2.2): Data ingestion, cleaning, transformation, and CSV I/O operations across all four datasets.
- scikit-learn (v1.4.2): TF-IDF vectorisation, model training (RandomForestClassifier, LogisticRegression, CalibratedClassifierCV wrapping LinearSVC), Pipeline construction, train-test splitting with stratification, and accuracy evaluation including top-k accuracy scoring.
- NumPy (v1.26.4): Array operations and np.argsort() for ranked retrieval of top-N career predictions by probability.
- FastAPI (v0.111.0) and Uvicorn (v0.29.0): REST API framework and ASGI server providing eight HTTP endpoints with automatic OpenAPI documentation generation and Pydantic-based input validation.
- pickle (built-in): Serialisation and deserialisation of the trained scikit-learn Pipeline (career_model.pkl, ~2 MB) and LabelEncoder (label_encoder.pkl, ~5 KB).

The TF-IDF vectoriser was configured with ngram_range=(1,2) to capture both unigram and bigram features (e.g., 'machine learning' as a single feature), max_features=5000 to limit feature dimensionality, and sublinear_tf=True to apply logarithmic term-frequency scaling. Model selection was performed by training all three classifiers within scikit-learn Pipelines on the training split and evaluating them on the held-out test split under identical conditions. The best-performing model was serialised and deployed within the FastAPI application, which exposes the /predict endpoint as the primary inference interface.

3. RESULTS AND DISCUSSION

3.1 Results / Data Presentation

Table 2 presents the comparative classification performance of all three models evaluated on the held-out test set (20% of the labelled corpus, 86 records stratified by career class).

Table 2: Comparative Model Performance

Model	Top-1 Accuracy	Top-3 Accuracy	Top-5 Accuracy
Random Forest (n=300 trees)	39.5%	61.6%	72.1%
Logistic Regression ★ BEST	45.3%	65.1%	77.9%
Linear SVC (C=5.0, calibrated)	39.5%	57.0%	65.1%
Random Baseline (1/22 classes)	4.5%	13.6%	22.7%

Table 3 presents the per-career classification report for the selected Logistic Regression model, including precision, recall, and F1-score for each of the 22 career categories.

Table 3: Per-Career Classification Report — Logistic Regression

Career Category	Precision	Recall	F1-Score	Test Samples
Business Analyst	0.00	0.00	0.00	4
Civil Engineer	1.00	0.50	0.67	4
Content Writer	0.33	0.50	0.40	2
Data Analyst	0.33	0.33	0.33	3
Data Engineer	0.00	0.00	0.00	2
Data Scientist	0.00	0.00	0.00	3
DevOps Engineer	0.00	0.00	0.00	1
Digital Marketer	0.00	0.00	0.00	3
Embedded Systems Engineer	0.00	0.00	0.00	2
Financial Analyst	1.00	0.50	0.67	2
Graphic Designer	0.50	0.50	0.50	2
HR	0.00	0.00	0.00	2
ML/AI Engineer	0.50	0.60	0.55	5
Mechanical Engineer	0.58	0.78	0.67	9
Medical Practitioner	1.00	0.50	0.67	2
QA/Test Engineer	0.00	0.00	0.00	2
Research Scientist	0.00	0.00	0.00	2
Sales Executive	0.50	0.67	0.57	3
Software Developer	0.45	0.68	0.55	22
Teacher/Professor	0.50	0.50	0.50	6
UI/UX Designer	0.20	0.50	0.29	2
Web Developer	0.50	0.33	0.40	3

Table 4 summarises the end-to-end API response time breakdown for a representative /predict request.

Table 4: API Performance Profile — /predict Endpoint

Processing Step	Time (approx.)	Explanation
Input text cleaning (regex)	< 0.1 ms	Simple substitution on short string
TF-IDF vectorisation	1–3 ms	Sparse matrix multiplication (5000 features)
Logistic Regression predict_proba()	1–3 ms	Single matrix multiply: 22×5000 weight matrix
np.argsort() on 22 probabilities	< 0.1 ms	Trivially fast on 22 elements
College DataFrame filter (pandas)	1–5 ms	Scanning 470 rows in memory
Job DataFrame filter (pandas)	2–5 ms	Scanning 845 rows in memory
JSON serialisation (Pydantic)	1–3 ms	Rust-accelerated core serialiser
HTTP overhead (Uvicorn)	5–10 ms	Network stack and request parsing
Total end-to-end	< 50 ms	Well within 200 ms UX threshold

3.2 Result Analysis

Logistic Regression outperformed both Random Forest and Linear SVC on all three evaluation metrics. The superiority of Logistic Regression in this setting is attributable to the high-dimensional, sparse nature of TF-IDF feature vectors: in such spaces, linear classifiers with calibrated decision boundaries tend to generalise better than ensemble methods that benefit from feature interactions present in denser representations. The Random Forest classifier, despite its ensemble nature, scored identically to Linear SVC at Top-1 (39.5%), suggesting that the additional variance reduction offered by bagging does not compensate for the limited training signal available at this dataset size.

The per-career analysis in Table 3 reveals a pronounced performance disparity across classes. Careers with well-defined, domain-specific vocabulary — Civil Engineer, Financial Analyst, Medical Practitioner, ML/AI Engineer, Mechanical Engineer — achieved F1-scores of 0.55 to 0.67, reflecting the model's ability to exploit discriminative lexical signals (e.g., 'AutoCAD', 'MATLAB', 'Solidworks' for Mechanical Engineer; 'TensorFlow', 'neural network', 'deep learning' for ML/AI Engineer). In contrast, careers such as Business Analyst, Data Scientist, DevOps Engineer, and HR yielded zero recall on the test set, indicating that the model defaulted to more frequently occurring classes when encountering ambiguous inputs. This is a classic manifestation of class imbalance: Software Developer, with 22 test samples, dominated the test set, and the classifier's tendency to predict this class explains the non-trivial Top-1 accuracy despite the zero-recall classes.

The API performance analysis confirms that the system meets the sub-50 millisecond response target for all tested inputs. The dominant latency contributors are HTTP overhead from Uvicorn (5–10 ms) and pandas DataFrame filtering (3–10 ms combined), both of which are amenable to further optimisation through connection pooling and indexed lookups respectively.

3.3 Discussion

The results align with the theoretical expectations for text classification on small, noisy, multi-class datasets. The 10-fold improvement over random chance (45.3% vs. 4.5% Top-1 accuracy) confirms that the TF-IDF + Logistic Regression pipeline successfully captures meaningful associations between skill-interest profiles and career outcomes. The Top-3 accuracy of 65.1% is particularly relevant given the system's user-facing design:

students receive three ranked career recommendations, and the correct career appears among them in approximately two out of three cases.

The integration of NIRF-based institutional data and structured job listings substantially enhances the system's practical utility beyond what the classification accuracy alone suggests. A student who receives three career recommendations can immediately access a list of NIRF-ranked institutions relevant to each career, filtered by their academic score, budget, and preferred study mode, alongside a curated list of entry-level employment opportunities. This tightly coupled recommendation-to-action pipeline addresses a critical gap identified in the literature: most prior systems terminate at career prediction without providing the institutional and employment context necessary for actionable decision-making.

The choice of Logistic Regression over more complex architectures reflects a deliberate design principle: interpretability and deployability in resource-constrained environments. The complete inference stack runs on a standard laptop CPU, requires no GPU, and loads in under one second at API startup. This makes the system viable for deployment in rural and semi-urban educational institutions where computational resources are limited, directly targeting the accessibility gap identified in the background section.

4. CONCLUSION

4.1 Objective Review

This research set out to design, implement, and evaluate a complete, end-to-end AI-Based Career Guidance System that accepts natural-language student profiles and returns personalised career, institutional, and employment recommendations. All four sub-objectives were fulfilled: a high-quality 741-record training corpus was constructed from two heterogeneous public datasets; three machine learning classifiers were rigorously evaluated under identical experimental conditions; NIRF 2024 and fresher job data were integrated with multi-attribute filtering; and the system was deployed as a REST API achieving sub-50 millisecond inference latency.

4.2 Key Findings

- Logistic Regression with TF-IDF (1,2)-gram features achieved the highest classification performance among the three evaluated models, with Top-1, Top-3, and Top-5 accuracies of 45.3%, 65.1%, and 77.9% respectively — a tenfold improvement over the 4.5% random baseline across 22 career categories.
- Career classes with domain-specific, discriminative lexica (Mechanical Engineer, ML/AI Engineer, Financial Analyst) achieve substantially higher per-class F1-scores than semantically diffuse classes (Business Analyst, HR), underscoring the importance of dataset size and label specificity in multi-class text classification.
- Data preprocessing — specifically career label consolidation from 381 to 22 categories and the exclusion of non-working respondents from the survey dataset — was the single most impactful intervention for model performance, enabling the system to learn from an effective average of 15.6 training examples per class rather than fewer than 2.
- The FastAPI/Uvicorn REST API deployment achieves end-to-end response times well below 50 milliseconds, confirming the viability of lightweight ML architectures for real-time interactive guidance applications without GPU acceleration.

4.3 Implications / Applications

The system has direct applicability in secondary and higher-education guidance contexts within India. University career development cells can deploy the REST API as a back-end service for student-facing web portals, enabling self-service career exploration at scale. The NIRF-integrated institutional matching is particularly valuable for students from non-metropolitan regions who lack access to professional career counsellors. Government-sponsored digital education initiatives (e.g., SWAYAM, DIKSHA) could incorporate the system to complement existing course recommendation functionality with career pathway guidance.

From a research perspective, the fully open and reproducible pipeline — from raw CSV ingestion through API deployment — provides a replicable baseline against which future improvements in feature engineering, model architecture, and dataset expansion can be benchmarked. The per-class diagnostic report in Table 3 explicitly identifies which career categories require additional training data, providing a targeted roadmap for dataset augmentation efforts.

4.4 Future Work

Several directions present themselves as natural extensions of the present work. First, expanding the training corpus through active collection of labelled career profiles from Indian educational institutions and professional networks would directly address the data sparsity limitation identified as the primary ceiling on current performance. Targeting a minimum of 200 labelled examples per career category is projected to raise Top-1 accuracy above 70% based on learning curve extrapolation.

Second, the replacement of TF-IDF with contextual sentence embeddings generated by multilingual transformer models (e.g., MuRIL, IndicBERT) would enable the system to process skill descriptions entered in Hindi and other regional languages, substantially broadening its accessibility to non-English-speaking student populations.

Third, the integration of real-time job market data via APIs from major Indian employment platforms (Naukri, LinkedIn, Internshala) would ensure that employment recommendations reflect current hiring trends rather than a static snapshot, significantly increasing their relevance to graduating students.

Finally, the incorporation of an explicit student feedback loop — where users rate the relevance of received recommendations — would enable online learning and reinforcement from user interactions, creating a self-improving system that adapts to evolving career landscapes without requiring periodic full retraining.

REFERENCES

- [1] Aspiring Minds. (2019). National Employability Report — Engineering Graduates. Aspiring Minds Research. Retrieved from <https://www.aspiringminds.com/research/national-employability-report>.
- [2] Holland, J. L. (1997). Making Vocational Choices: A Theory of Vocational Personalities and Work Environments (3rd ed.). Psychological Assessment Resources.
- [3] Reschly, A. L., & Christenson, S. L. (2006). Prediction of Dropout Among Students with Mild Disabilities. Remedial and Special Education, 27(5), 276–292.
- [4] Gupta, A., Mishra, S., & Tripathi, A. (2020). Career Recommendation System Using Collaborative Filtering and Matrix Factorisation. Proceedings of the International Conference on Advances in Computing and Data Sciences (ICACDS), Springer, 512–521.
- [5] Kumar, R., Sharma, P., & Jain, V. (2021). Career Prediction Using Naive Bayes Classifier on Student Survey Data. International Journal of Innovative Technology and Exploring Engineering, 10(4), 88–94.
- [6] Jain, D., & Sharma, N. (2022). Multi-Class Career Prediction Using Support Vector Machines and TF-IDF Features. Journal of Emerging Technologies and Innovative Research, 9(7), f112–f119.
- [7] Nguyen, T., Pham, H., & Le, Q. (2023). BERT-Based Career Recommendation for University Students. Proceedings of the IEEE International Conference on Educational Technology, 234–241.
- [8] Roy, S., & Banerjee, A. (2022). A Hybrid Career Counselling System Integrating Skill-Based Prediction with Institutional Matching. International Journal of Computer Applications, 183(12), 22–29.
- [9] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- [10] Salton, G., & Buckley, C. (1988). Term-Weighting Approaches in Automatic Text Retrieval. Information Processing & Management, 24(5), 513–523.
- [11] Ramachandran, T., & Arun, K. (2023). FastAPI for Machine Learning Microservices: Design Patterns and Benchmarks. Journal of Open Source Software, 8(83), 5217.
- [12] Ministry of Education, Government of India. (2024). National Institutional Ranking Framework (NIRF) Rankings 2024. Retrieved from <https://www.nirfindia.org/Rankings/2024/EngineeringRanking.html>.
- [13] McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference (SciPy 2010), 51–56.
- [14] Harris, C. R., et al. (2020). Array Programming with NumPy. Nature, 585, 357–362.
- [15] Tibor, G., & Csaki, C. (2021). Comparative Analysis of Machine Learning Algorithms for Career Path Prediction. Procedia Computer Science, 181, 991–998.